



جامعة نيويورك أبوظبي
NYU ABU DHABI

Advanced Circuits

ENGR-UH 2311 | Spring I - 2025

Experiment: **Lab 4**

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Lab 4: Transient Simulation of Circuits Involving RLC in Cadence and Sources with Varying Phases

Objectives

The main objective of this lab is to get familiar with the Cadence environment, create schematics, define component parameters, and configure various voltage and current sources. It includes setting up and running transient simulations to analyze circuit behaviors, observe phase differences, and generate high-resolution schematics and plots for analysis.

Diagram

Circuit 1:

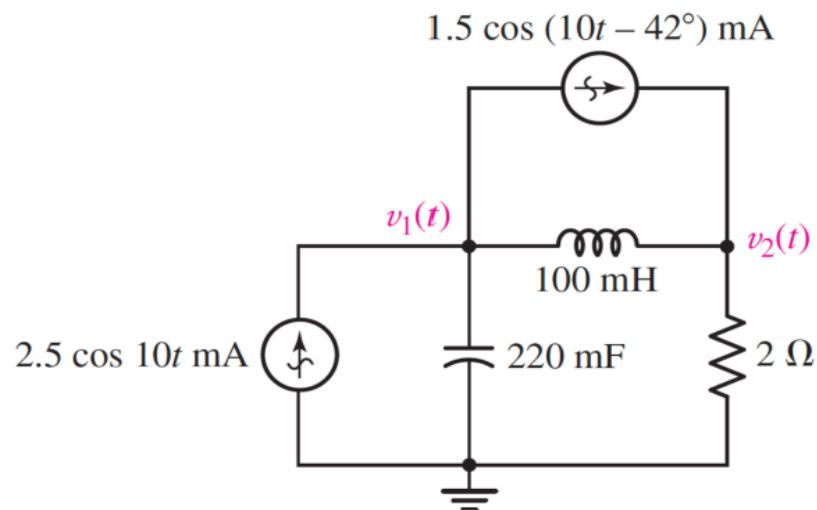


Figure 1: Circuit 1 Schematic

Circuit 2:

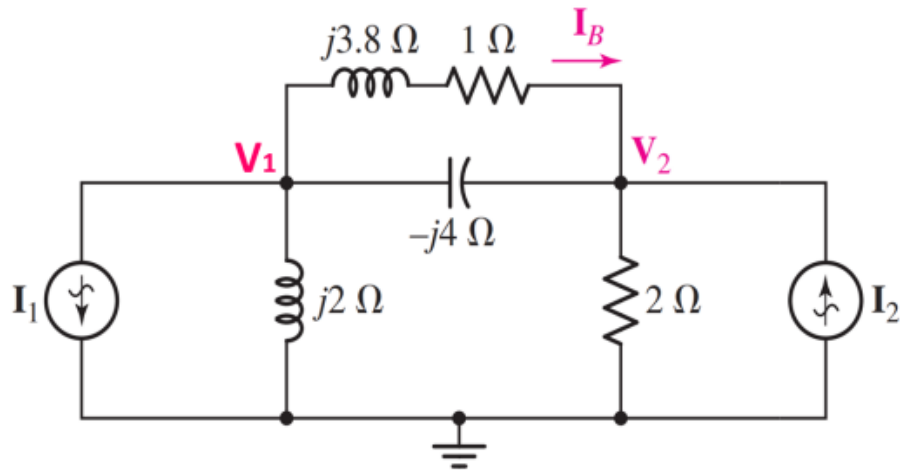


Figure 2: Circuit Schematic 2

Procedure

Circuit 1: RLC Circuit with Phase Differences

1. An RLC circuit was built in Cadence, consisting of a resistor ($R = 2\Omega$), an inductor ($L = 100\text{mH}$), a capacitor ($C = 220\text{mF}$) and 2 current sources as shown in the schematic in Figure 1.
2. Angular frequency $\omega = 10 \text{ rad/s}$ was used to find the frequency of $10/6.28 \text{ Hz}$.
3. Current source 1: $2.5 \cos(10t) \text{ mA}$ was converted to its sine equivalent of $2.5 \sin(10t + 90) \text{ mA}$ and was configured appropriately to create phase differences as required.

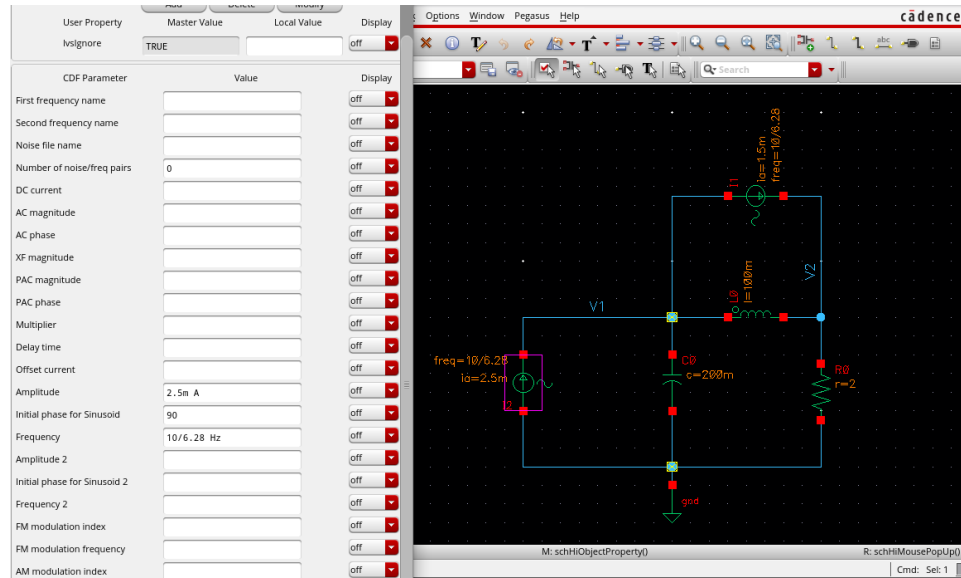


Figure 3: Configuration of Current Source 1 (2.5)

4. Current source 2: $1.5 \cos(10t-42)$ mA was converted to its sine equivalent of $1.5 \sin(10t + 48)$ mA and was configured appropriately to create phase differences as required.

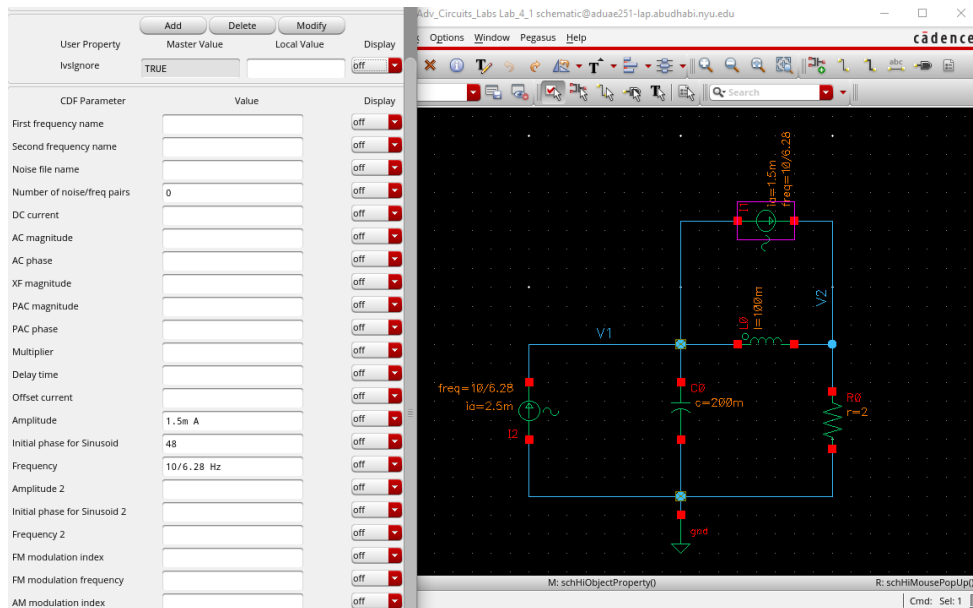


Figure 4: Configuration of Current Source 2 (1.5)

5. The transient simulation was set up to measure the steady-state behavior of mesh currents.
6. The following plots were recorded and analyzed:
 - $v_1(t)$ and current through the capacitor, with estimated phase difference.
 - $v_2(t)$ and current through the resistor, with estimated phase difference.

- $(v_2 - v_1)$, transient voltage difference across the inductor, and current through the inductor, with estimated phase difference.
7. The simulation results were examined to verify theoretical expectations.

Circuit 2: RLC Circuit

1. An RLC circuit with a switch was implemented in Cadence, consisting of 2 resistors ($R = 1\Omega$ and $R = 2\Omega$ each), 2 inductors ($L = j3.8\Omega$ and $L = j2\Omega$ each), a capacitor ($C = -j4\Omega$), and 2 current sources as shown in Figure 2.
2. Angular frequency $\omega = 1.2$ rad/s was used to find the frequency of 1.2/6.28 Hz.
3. Current source 1: $5\angle-18^\circ$ A was converted to its sine equivalent of $5 \sin(1.2t + 72^\circ)$ A and was configured appropriately to create phase differences as required.

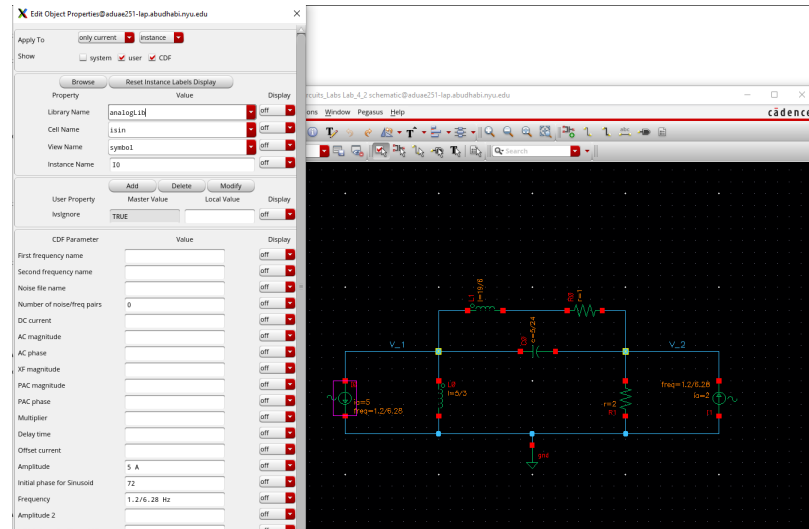


Figure 5: Configuration of Current Source 1 (5)

4. Current source 2: $2\angle5^\circ$ A was converted to its sine equivalent of $2 \sin(1.2t + 95^\circ)$ A and was configured appropriately to create phase differences as required.

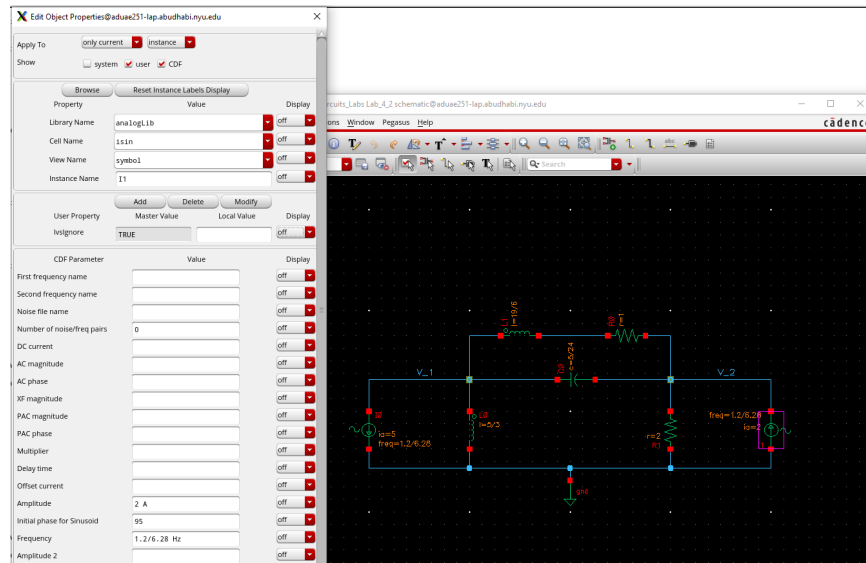


Figure 6: Configuration of Current Source 2 (2)

5. The transient analysis was performed with the given angular frequency and currents.
6. The following plots were recorded for analysis:
 - Plot of V_2 vs. I_B , with estimated phase difference.
 - Plot of V_1 vs. current through the inductor ($j2\Omega$), with estimated phase difference.
 - Plot of $(V_2 - V_1)$ vs. current through the capacitor ($-j4\Omega$), with estimated phase difference.
7. The simulated outcomes were compared with theoretical calculations to ensure accuracy.

Results & Observations

Circuit 1

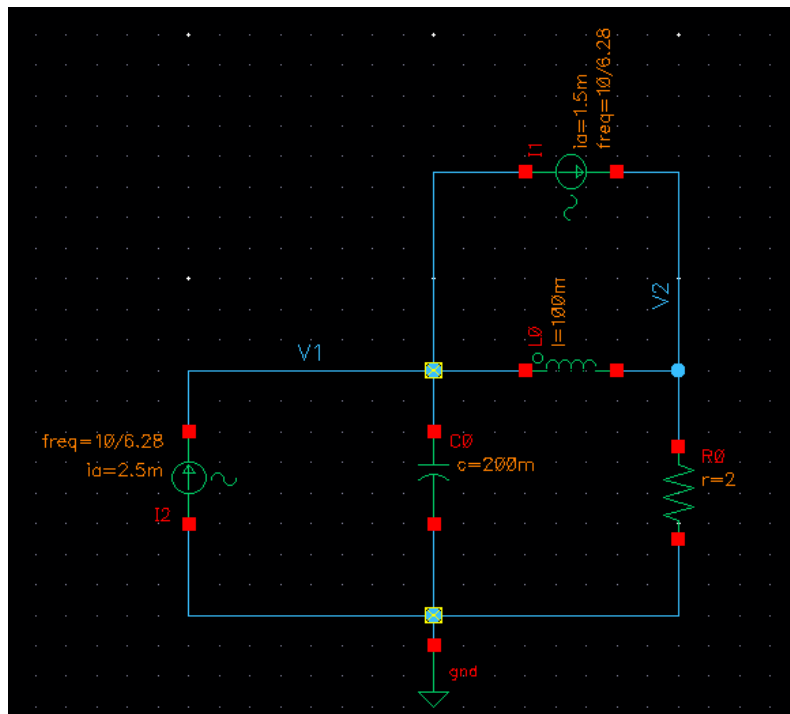
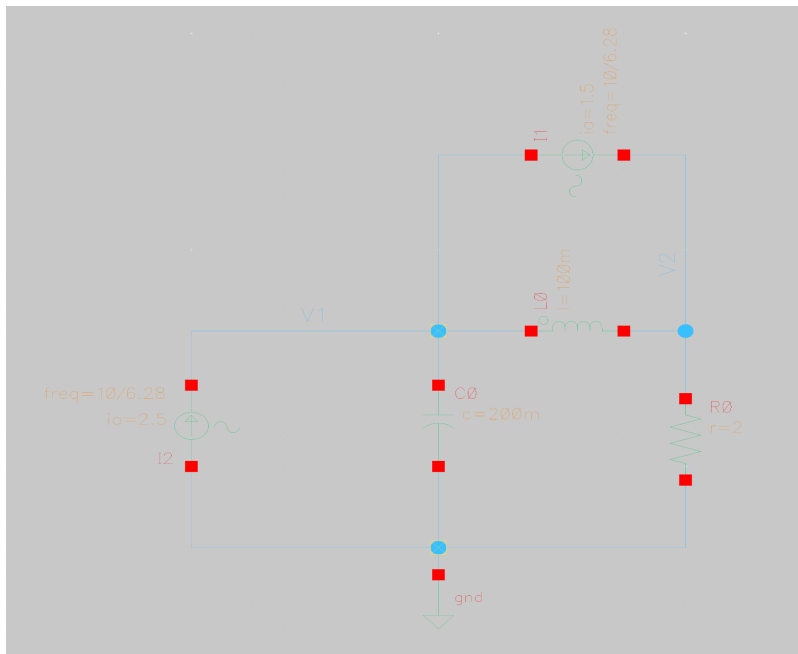


Figure 7: Circuit Schematic 1 Cadence Implementation

Plots Obtained For Circuit Schematic 1

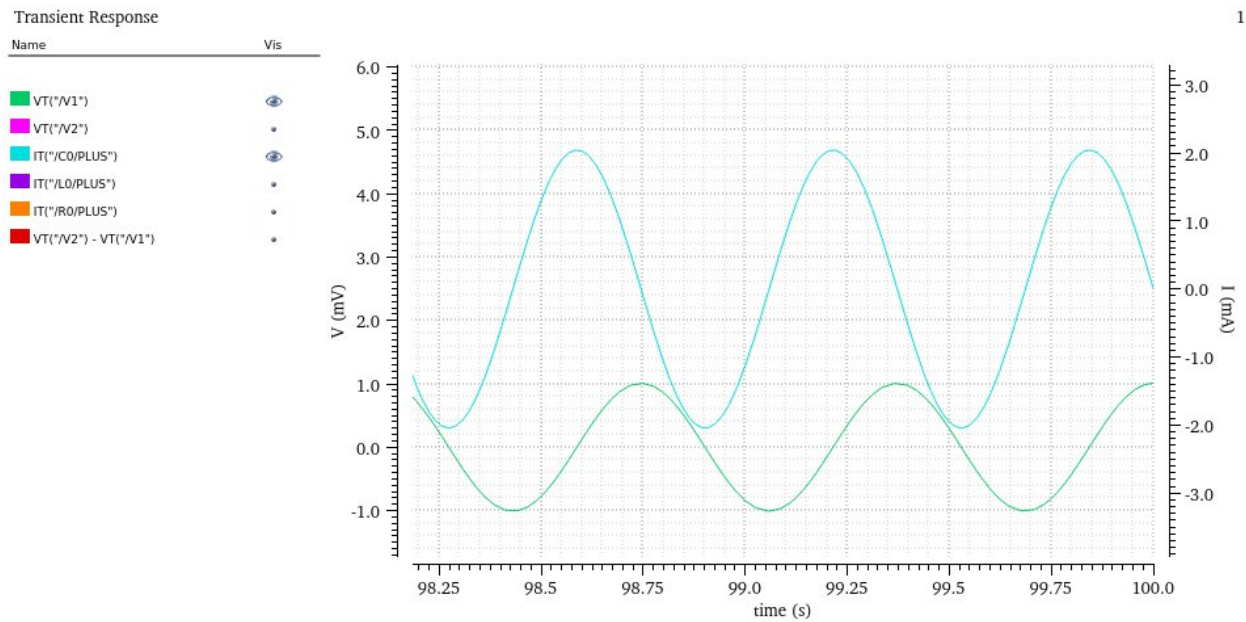


Figure 8: Current and Voltage Plots of Capacitor

The graph in Figure 4 above plots the voltage (green) and current (blue) of the capacitor in RLC Circuit 1. A phase difference of approximately 90° is observed with the current leading the voltage.

Transient Response

1

Name	Vis
VT("V1")	☐
VT("V2")	☒
IT("C0/PLUS")	☐
IT("L0/PLUS")	☐
IT("R0/PLUS")	☒
VT("V2") - VT("V1")	☐

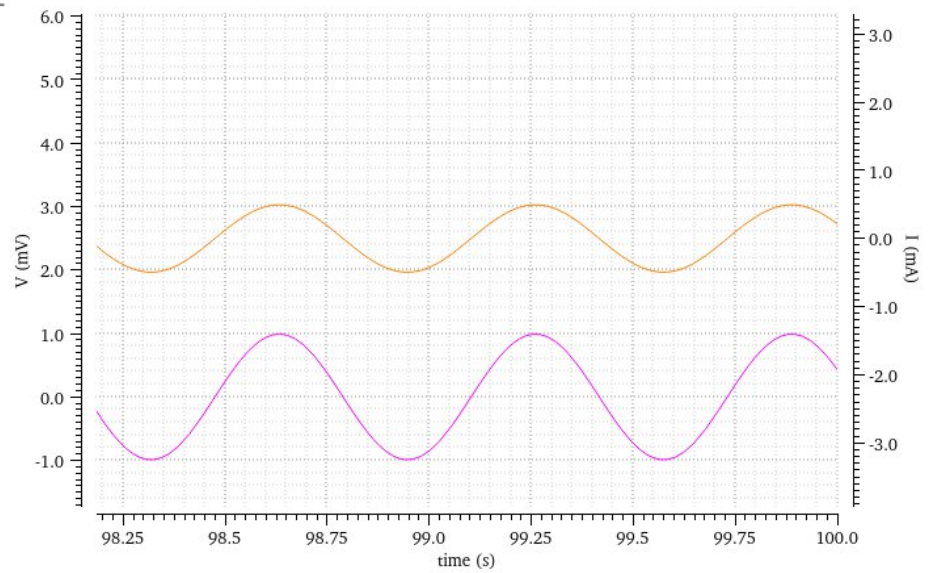


Figure 9: Current and Voltage Plots of Resistor

The graph in Figure 5 above plots the voltage (pink) and current (orange) of the resistor in RLC Circuit 1. A phase difference of approximately 0° is observed with the current and voltage in phase.

Transient Response

1

Name	Vis
VT("V1")	☐
VT("V2")	☐
IT("C0/PLUS")	☐
IT("L0/PLUS")	☒
IT("R0/PLUS")	☐
VT("V2") - VT("V1")	☒

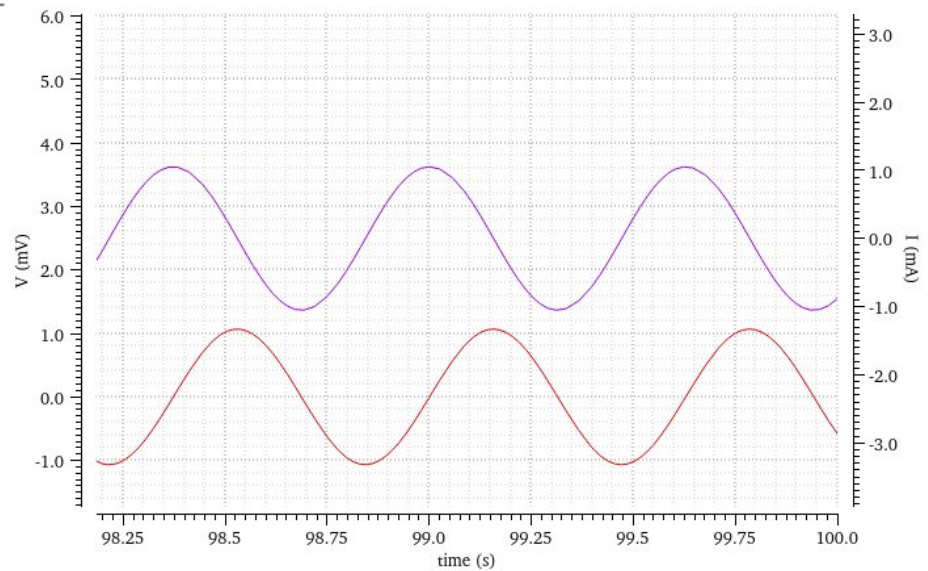


Figure 10: Current and Voltage Plots of Inductor

The graph in Figure 6 above plots the voltage (red) and current (purple) of the inductor in RLC Circuit 1. A phase difference of approximately 90° is observed with the current leading the voltage.

Circuit 2

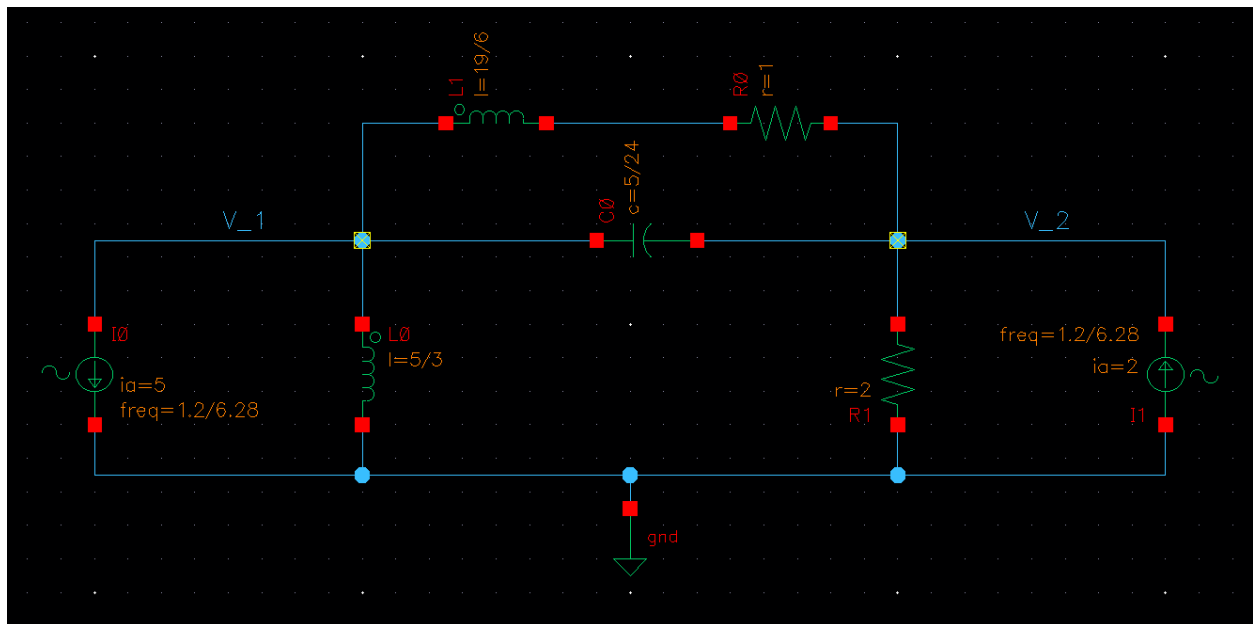
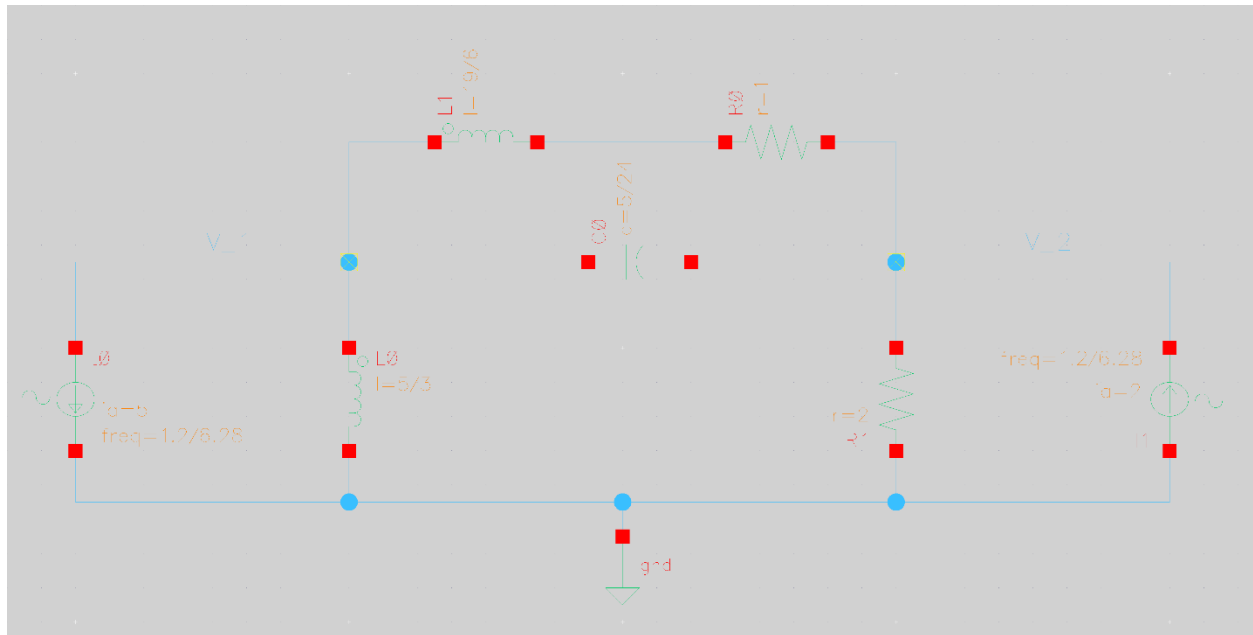


Figure 11: Circuit Schematic 2 Cadence Implementation

Plots Obtained For Circuit Schematic 2

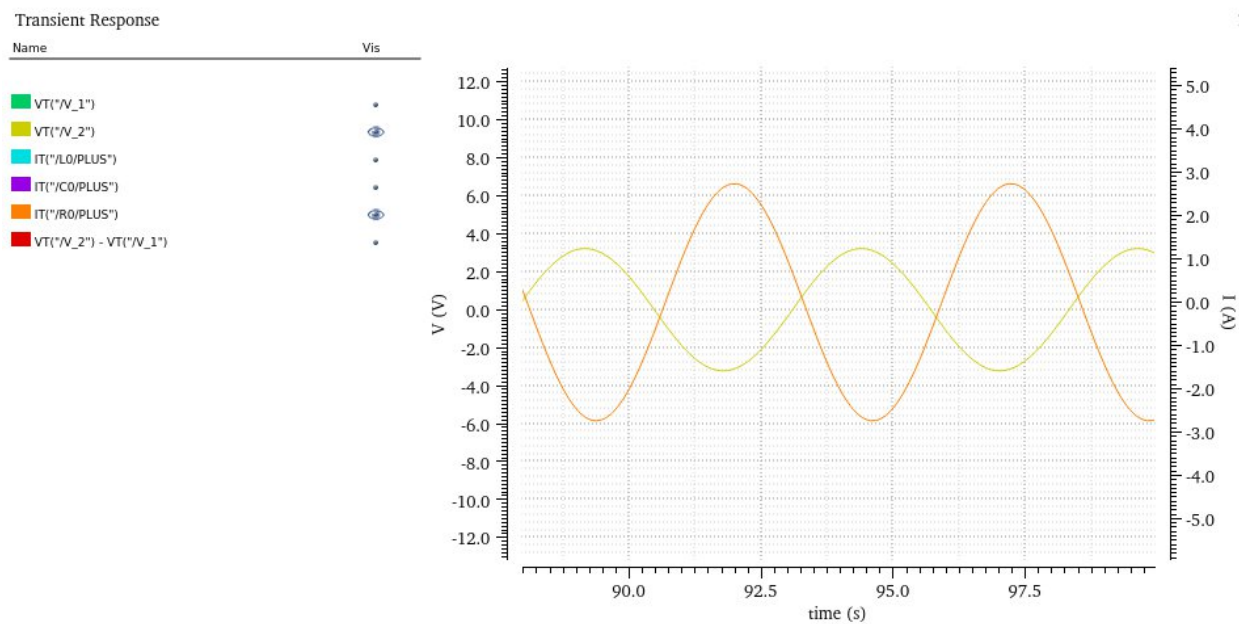


Figure 12: Plots of Current I_B and Voltage V_2

The graph in Figure 8 above plots the voltage (yellow) and current (orange) of the node V_2 and the current I_B in RLC Circuit 2. A phase difference of approximately 180° is observed with the voltage leading the current.

Transient Response

Name	Vis
VT("V_1")	☒
VT("V_2")	☐
IT("L0/PLUS")	☒
IT("C0/PLUS")	☐
IT("R0/PLUS")	☐
VT("V_2") - VT("V_1")	☐

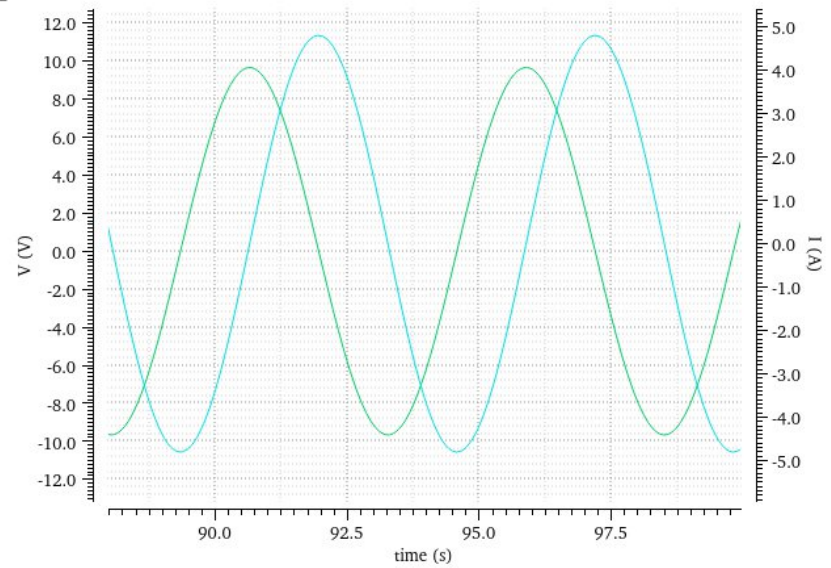


Figure 13: Current and Voltage Plots of Inductor $j2\ \Omega$

The graph in Figure 9 above plots the voltage (green) and current (blue) of the inductor $j2\ \Omega$ in RLC Circuit 2. A phase difference of approximately 90° is observed with the voltage leading the current.

Transient Response

1

Name	Vis
VT("V_1")	•
VT("V_2")	•
IT("L0/PLUS")	•
IT("C0/PLUS")	•
IT("R0/PLUS")	•
VT("V_2") - VT("V_1")	•

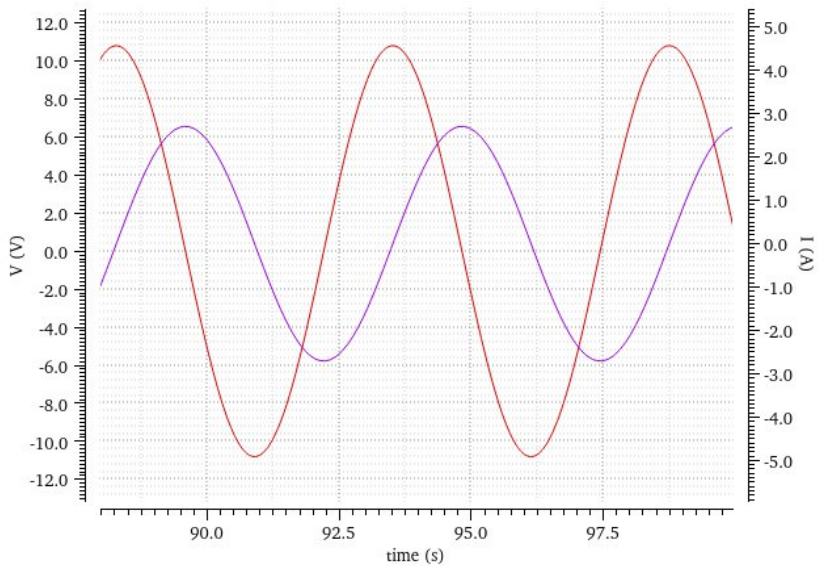


Figure 14: Current and Voltage Plots of Inductor $-j4 \Omega$

The graph in Figure 10 above plots the voltage (red) and current (purple) of the capacitor $-j4 \Omega$ in RLC Circuit 2. A phase difference of approximately 90° is observed with the voltage leading the current.

Conclusion

The results of this lab reinforced our understanding of transient responses in RLC circuits by accurately simulating phase differences and oscillatory behavior using Cadence. The observed waveforms aligned well with theoretical predictions, confirming the validity of our setup and configurations. Through analysis of voltage and current plots, we were able to visualize how impedance elements influence transient behavior and steady-state conditions. This experiment provided valuable insights into designing and analyzing electrical circuits with phase-shifted sources, which are essential for real-world engineering applications.